

NAIROBI DAGORETTI, Kenya

WMO: 637410

Lat: 1.3013S

Lon: 36.7597E

Elev: 1798

StdP: 81.51

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 10.0	(c) 11.1	(d) 3.6	(e) 6.1	(f) 24.8	(g) 5.2	(h) 6.8	(i) 23.4	(j) 5.2	(k) 20.3	(l) 4.5	(m) 20.0	(n) 0.1	(o) 160	(p) 0.317

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 3	(b) 10.6	(c) 27.6	(d) 15.0	(e) 26.9	(f) 15.1	(g) 26.1	(h) 15.3	(i) 18.3	(j) 22.5	(k) 18.0	(l) 22.2	(m) 17.6	(n) 21.8	(o) 4.1	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	15.3	19.2	16.8	15.0	18.9	16.4	14.6	18.4	60.1	22.6	58.5	22.1	57.3	21.7	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years	
(a)	(b)	(c)		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
6.4	5.6	5.1	DB	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
			WB	8.0	29.5	1.1	1.4	7.2	30.5	6.5	31.3	5.9	32.0	5.1	33.0	
				7.1	21.0	1.2	1.4	6.2	22.0	5.5	22.8	4.8	23.6	3.9	24.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.9	19.5	20.2	20.6	20.1	18.7	17.5	16.7	17.1	18.6	19.5	19.0	19.0
	DBStd	1.66	1.06	1.06	1.22	1.11	1.01	1.21	1.35	1.28	1.48	1.21	1.17	0.94
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	164	3	1	1	1	8	30	52	42	14	4	6	4
	CDD10.0	3233	293	284	327	302	270	226	209	220	258	293	271	279
	CDD18.3	355	38	52	69	53	19	6	2	3	22	39	27	25
	CDH23.3	1530	246	363	382	127	19	16	18	27	118	137	40	38
	CDH26.7	81	7	27	32	7	0	0	0	0	4	3	1	0
Wind	WSAvg	1.7	2.3	2.2	2.5	1.7	1.1	0.8	0.8	1.0	1.6	2.2	2.2	2.4
Precipitation	PrecAvg	1043	54	47	95	245	189	38	17	24	30	59	148	98
	PrecMax	1652	199	182	299	707	532	141	69	77	138	196	623	379
	PrecMin	483	0	0	6	20	3	4	1	3	0	2	21	8
	PrecStd	282	56	51	70	152	120	32	16	21	29	49	108	83
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.4	28.2	28.6	27.5	25.0	25.1	25.0	25.2	27.1	27.1	25.9	25.2
		MCWB	14.5	14.6	14.8	15.9	16.9	16.4	15.8	15.5	15.0	15.3	15.8	16.2
	2%	DB	26.8	27.4	27.7	25.8	23.9	23.2	23.6	24.1	25.8	26.0	24.4	24.3
		MCWB	14.6	14.7	15.2	16.3	17.0	16.5	15.6	15.5	15.2	15.3	16.3	16.2
	5%	DB	25.8	26.8	26.7	24.8	23.0	22.1	22.3	23.0	24.8	25.0	23.6	23.6
		MCWB	15.0	14.8	15.4	16.7	16.7	16.1	15.2	15.0	15.1	15.4	16.3	16.2
	10%	DB	24.9	25.9	25.8	23.8	22.0	21.1	21.1	21.8	23.7	23.9	22.7	22.8
		MCWB	15.2	15.1	15.6	16.8	16.5	15.8	15.0	14.9	15.1	15.5	16.3	16.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.1	18.7	18.4	18.9	18.6	18.2	17.2	17.2	17.5	17.7	18.2	18.2
		MCDB	23.3	23.1	23.7	22.7	22.3	21.9	22.2	22.1	23.0	22.0	22.3	21.8
	2%	WB	17.2	18.0	17.7	18.2	17.9	17.3	16.5	16.5	16.6	17.1	17.5	17.6
		MCDB	22.5	22.8	22.8	22.1	21.7	21.1	21.4	21.7	22.3	21.6	21.5	21.2
	5%	WB	16.7	17.2	17.2	17.8	17.4	16.8	15.8	15.7	16.1	16.7	17.1	17.2
		MCDB	22.0	21.9	22.1	21.7	20.9	20.5	20.7	21.0	22.0	21.0	20.9	20.9
	10%	WB	16.3	16.8	16.9	17.4	17.0	16.4	15.3	15.3	15.6	16.4	16.7	16.8
		MCDB	21.5	21.5	21.7	21.2	20.3	20.0	20.0	20.1	21.3	20.5	20.3	20.5
Mean Daily Temperature Range	5% DB	MDBR	11.6	12.0	10.6	8.6	7.4	8.1	8.9	9.2	10.6	9.3	7.9	8.2
		MCDBR	13.1	13.8	12.3	9.8	8.8	9.8	11.5	12.1	12.4	11.1	9.1	9.3
		MCWBR	4.1	4.1	3.3	2.9	2.8	4.1	4.8	4.6	3.9	3.0	2.6	3.0
	5% WB	MCDBR	10.5	10.3	9.8	8.5	7.5	8.8	10.1	10.3	10.9	8.5	8.0	7.9
		MCWBR	3.8	3.9	3.2	3.0	2.9	3.9	4.6	4.4	4.0	2.9	2.9	2.9
Clear-Sky Solar Irradiance	taub		0.362	0.390	0.373	0.321	0.320	0.356	0.370	0.357	0.338	0.350	0.344	0.355
	taud		2.453	2.354	2.432	2.646	2.640	2.483	2.403	2.434	2.495	2.464	2.520	2.492
	Ebn at Noon		969	942	949	977	955	905	897	930	968	969	980	972
	Edh at Noon		118	132	121	95	92	105	115	116	112	116	109	112
All-Sky Solar Radiation	RadAvg		6.48	6.78	6.58	5.80	5.31	4.73	4.69	4.98	5.97	6.07	5.63	6.00
	RadStd		0.42	0.35	0.43	0.31	0.20	0.32	0.44	0.36	0.33	0.30	0.31	0.37

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	+0.47	+0.69	N/A	+0.57	+0.39	+0.16	N/A	N/A	N/A	N/A

Nomenclature: See separate page